Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
Alangilan Campus
Golden Country Homes, Alangilan Batangas City, Batangas, Philippines 4200
Tel Nos.: (+63 43) 425-0139 local 2121 / 2221
E-mail Address: coe.alangilan@g.batstate-u.edu.ph | Website Address: http://www.batstate-u.edu.ph

College of Engineering

Generated Program of Study

TERMINAL USER | N/A | A.Y. N/A | Evaluated: April 02, 2026 | Solver: cp_sat (OR-Tools CP-SAT) | config=config1_baseline

1st Year 2nd Sem — 21 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
ENGG 102	Engineering Drawing	1	0	3			
ENGG 114	Basic Occupational Safety and Health	3	3	0			
GEd 101	Understanding the Self	3	3	0			
GEd 102	Mathematics in the Modern World	3	3	0			
GEd 106	Purposive Communication	3	3	0			
GEd 109	Science, Technology and Society	3	3	0			
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1		

1st Year Summer — 6 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
GEd 103	Life and Works of Rizal	3	3	0			
GEd 107	Ethics	3	3	0			

2nd Year 1st Sem — 18 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
AI 101	Linear Algebra for AI	3	3	0			
CpE 201	CpE Discipline, Professional Practice and laws	3	3	0			
CpE 203	Programming Logic and Design	2	0	6			
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102		
GEdE 101	The Entrepreneurial Mind	3	3	0			
MATH 101	Differential Calculus	4	4	0			
PATHFit 3	Choice of dance, sports, Martial Arts, Group exercise, Outdoor and Adventure Activities 1	2	2	0	PATHFit 1 & 2		

2nd Year 2nd Sem — 12 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
CpE 205	Object-Oriented Programming	2	0	6	CpE 203		
MATH 102	Integral Calculus	4	4	0	MATH 101		
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 2	2	2	0	PATHFit 1 & 2		
PHYS 102	Physics for Engineers 1	4	3	3	MATH 101	MATH 102	

2nd Year Summer — 3 units

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Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
PHYS 103	Physics for Engineers 2	3	2	3	PHYS 102; MATH 102		

3rd Year 1st Sem — 15 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
CpE 202	Discrete Mathematics	3	3	0	MATH 101		
CpE 303	Data Structures and Algorithms	2	0	6	CpE 205		
EE 212	Fundamentals of Electrical Circuits	4	3	3	PHY 103		
ENGG 104	Engineering Economics	3	3	0	MATH 102		
MATH 103	Differential Equations	3	3	0	MATH 102		

3rd Year 2nd Sem — 20 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		
CpE 204	Advance Engineering Mathematics for CpE	3	3	0	2nd Year Standing		
CpE 309	Software Design	4	3	3	CpE 303		
ECE 405	Electronic Circuits: Devices and Analysis	4	3	3	EE 212		
Elective Course 1	Elective Course 1	3					
MATH 104	Numerical Methods	3	3	0	MATH 103		

3rd Year Summer — 1 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
CpE 315	Computer Engineering Drafting and Design	1	0	3	ECE 405		

4th Year 1st Sem — 19 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
CpE 302	Logic Circuits and Design	4	3	3	ECE 405		
CpE 304	Introduction to Networks, Data and Digital Communications	3	2	3	ECE 405	CpE 302	
CpE 305	Fundamentals of Mixed Signals and Sensors	3	3	0	ECE 405		
CpE 306	Feedback and Control Systems	3	3	0	CpE 206; EE 212		
ENGG 116	Methods of Research	3	3	0	AI 102, 3rd Year Standing		
Elective Course 2	Elective Course 2	3					

4th Year 2nd Sem — 19 units

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Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
AI 103	Machine Learning and Neural Networks	3	2	3	AI 101, AI 102, CpE 101		
CpE 308	Microprocessors	4	3	3	CpE 302		
CpE 310	Digital Signal Processing	4	3	3	CpE 305; CpE 306		
CpE 311	Computer Architecture and Organization	4	3	3		CpE 308	
CpE 312	Connecting Networks and Security	4	3	3	CpE 304		

4th Year Summer — 4 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
CpE 313	Introduction to HDL	1	0	3	CpE 203, CpE 302		
CpE 314	Emerging Technologies in CpE	3	3	0	3rd Year Standing		

5th Year 1st Sem — 18 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
CpE 401	Operating Systems	3	3	0	CpE 303		
CpE 402	Embedded Systems	4	3	3	CpE 308		
CpE 403	Seminars and Fieldtrips	2	0	6	4th Year Standing		
CpE 404	CpE Practice and Design 1	1	0	3	ENGG 116, CpE 308		
ENGG 105	Technopreneurship	3	3	0	4th Year Standing		
ENGG 106	Engineering Management	2	2	0	4th Year Standing		
Elective Course 3	Elective Course 3	3					

5th Year 2nd Sem — 8 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade
CpE 405	CpE Practice and Design 2	2	0	6	CpE 404		
CpE 406	Comprehensive Examination for CpE	2	0	6	Graduating		
ENGG 117	OJT/Internship (320 hrs)	4	4	0	4th Year Standing		

5th Year Summer — 0 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade

6th Year 1st Sem — 0 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade

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6th Year 2nd Sem — 0 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade

6th Year Summer — 0 units

Course Code	Course Title	Units	Lec	Lab	Pre-requisite(s)	Co-requisite(s)	Grade